George Yazji

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Backend software engineer with 3+ years of experience building production systems. Co-founded a live shuttle-tracking platform serving 2,500+ weekly active users while owning enterprise features at SOTI deployed across ~100,000 devices.

WORK EXPERIENCE

Associate Software Developer May 2025 – Present
SOTI Inc. Mississauga, ON

- Implemented SQL Server Windows Authentication in SOTI Snap's Data Access Controller, adding **Python** backend authentication, error handling, service restart logic, and automatic re-authentication
- Led development of Scanner for Snap App forms, implementing **Android in Kotlin** and **C#/.NET** server support with Google ML Kit for live OCR and barcode capture, reducing data-entry time from **minutes to under 5 seconds**
- Owned end-to-end delivery of **5 production feature epics** for SOTI Snap, an enterprise low-code platform, spanning architecture, technical design, implementation, and cross-functional delivery across Android, server, and iOS
- Designed and built an on-device request-caching system, providing offline-first reliability across **~100,000 Snap devices** by persisting failed third-party API requests and automatically retrying them when connectivity returns

Lead Software Engineer & Co-Founder (Part-time) May 2025 – Present
Trac (tracinc.ca) Hamilton, ON

- Architected and implemented Trac's **Firebase** backend and real-time event pipeline, processing vehicle telemetry updates **every 5 seconds** and managing session state, ETA/occupancy updates, stale-vehicle detection, authentication, and data synchronization across commuter and driver clients
- Co-founded and launched Trac, serving **2,500+ weekly active users** at McMaster University through **Flutter**-based **iOS** and **Android** applications; secured a B2B operating partnership with Attridge Transportation
- Developed and deployed a multiple linear regression model using **Python** and **scikit-learn** for shuttle arrival prediction, achieving **~30-second average prediction error** from historical stop-level wait times and operational data
- Built real-time vehicle-processing algorithms using **Kalman filtering** to smooth noisy GPS data and **A*** to calculate stop-level routing distances for live shuttle tracking and ETA calculations

Software Engineering Intern May 2023 – August 2024
SOTI Inc. Mississauga, ON

- Engineered an internal **C#/.NET** diagnostic tool that uses binary search across batches of pull requests to identify breaking commits and notify responsible developers, reducing investigation time from **60 minutes to 10 minutes**
- Fixed flaky BDD/E2E tests and authored 150+ new tests, reducing failure frequency by **40%**
- Improved MobiControl reliability by resolving **20+** customer-reported defects and shipping production features across Java, Kotlin, and C#, including fixing a circular dependency in the Android codebase that caused concurrency issues

Undergraduate Research Assistant January 2023 – May 2023
McMaster Faculty of Engineering Hamilton, ON

- Designed and tuned a **PID controller** with **OpenCV**, improving RC car path-following from **60% to 95%**
- Implemented a Gipps car-following simulation on 4 Raspberry Pi vehicles via **MQTT**, maintaining **10 cm gaps**

PROJECTS

Autonomous Self-Parking Cars July 2024 – October 2024

- Built a **C++** pipeline in CARLA to generate and label **10,000+** bird's-eye-view training images
- Developed a ResNet-50 model in **PyTorch** for parking-space detection

SKILLS

Languages: C#, Kotlin, Java, Python, Dart, C++, SQL, JavaScript, TypeScript
Backend: .NET, REST APIs, SQL Server, Firebase (Firestore, Realtime Database, Authentication)
Mobile: Android, Flutter
AI/ML: PyTorch, OpenCV, scikit-learn, Google ML Kit
Tools: Git, Jenkins, Gerrit, CI/CD, Linux, CARLA, Android Studio, Xcode

EDUCATION

B.Eng. Software Engineering (Co-op), McMaster University | GPA 3.5/4.0, Dean's Honour List April 2025